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Scalable Personalized Medicine?

€ Unfortunately, still quite expensive

But why? Sequencing is getting cheap.

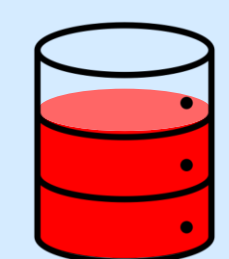
Answer:  Data privacy is crucial

 Genomic data storage is expensive

Question: Any way to fix that?

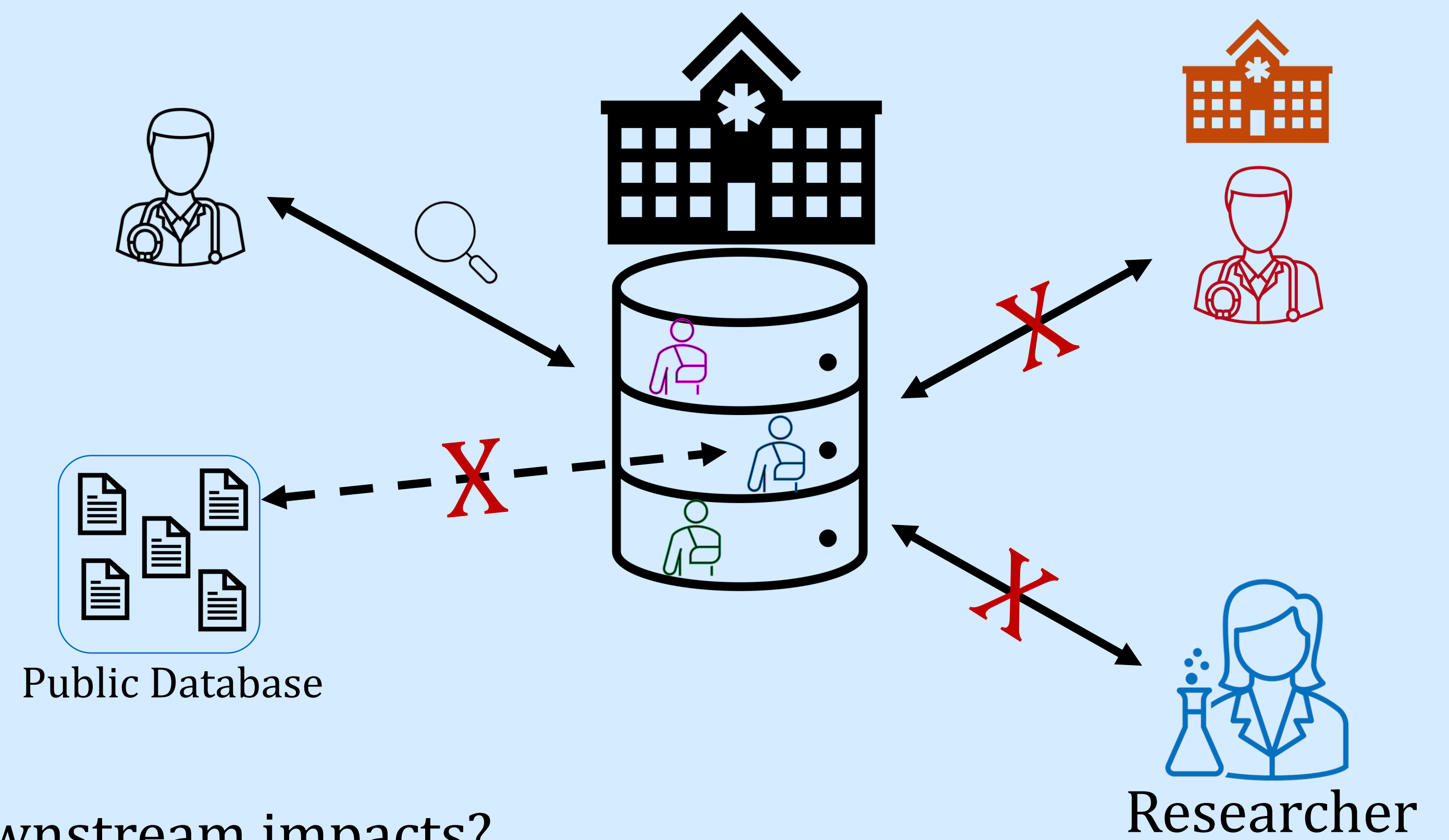
Personal Genome Sequence Data

 Variant Call Format (VCF)

 Very large (~GBs in size)

 Require strict privacy protections

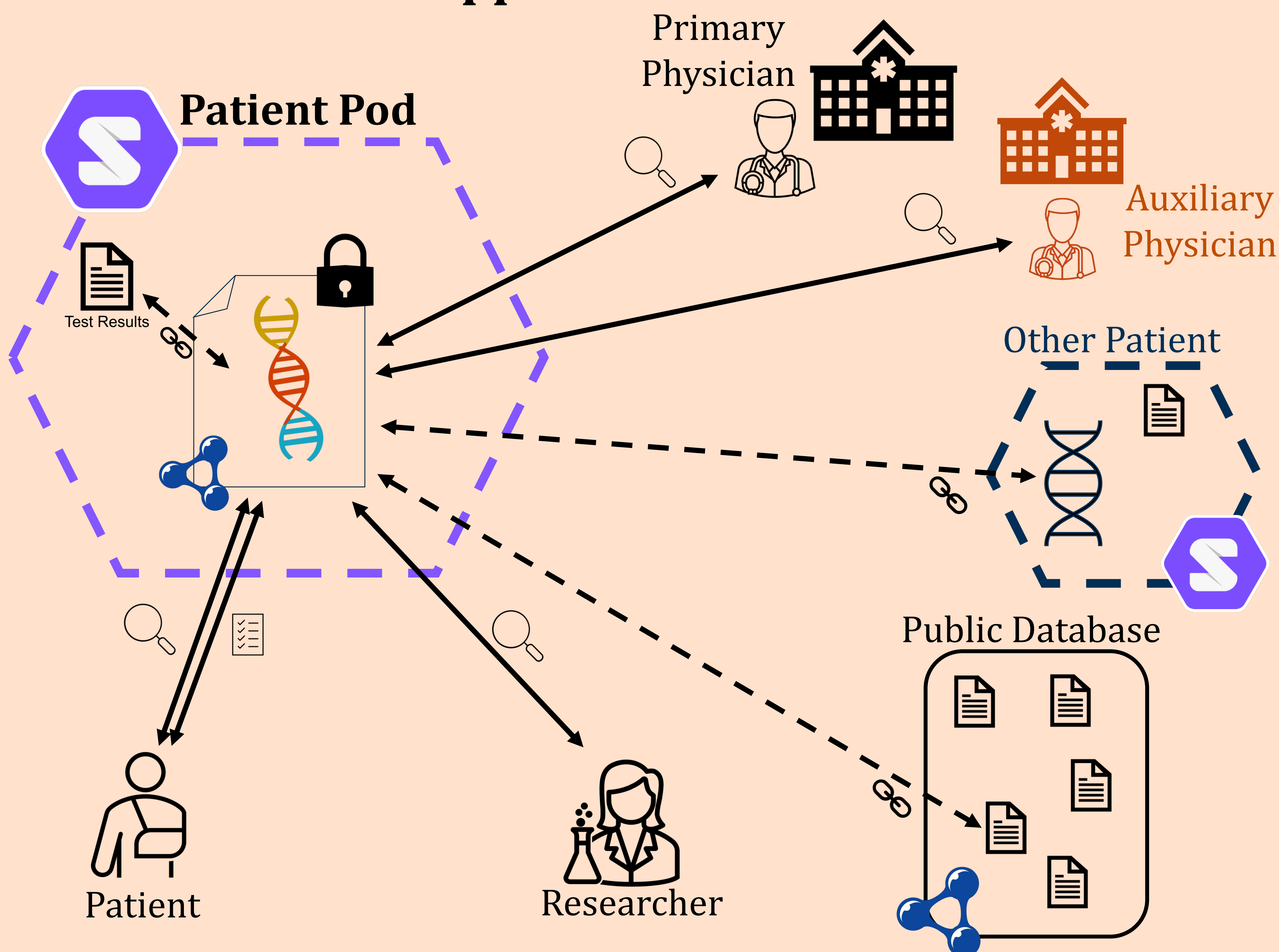
The current institution-centric system



Downstream impacts?

- Reduced data sharing
- Reduced patient transparency
- Increased pertinent data dispersion
- Data duplication

A Patient-centric Approach



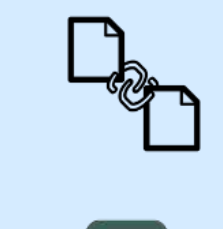
Key Challenges

-  Privacy granularity
 - ↳ Governance?
 - ↳ How granular is adequate?
-  RDF Conversion + Data Linkages
 - ↳ VCF to RDF Conversion?
 - ↳ Automation of data linking?
-  Querying performance
 - ↳ Many, large data sources
 - ↳ Algorithms that use indexes?

Engineering Tasks

-  Genomic data storage in Solid pods
-  Data privacy management
-  Data Sharing via the web

Research Pursuits

-  Represent genomic data as RDF?
-  Linking genomic and health data?
-  Single and Many pod Querying?